



2026

Chemistry

Paper 1 — Atomic Structure · Periodic Table · Ionic Bonding

Time allowed: 45 minutes

Total marks: 50

Index Number: _____

Instructions to Candidates

1. Write your Index Number in the space provided above.
2. Answer **all** questions in **all three sections**.
3. Section A: Choose **one** answer for each question. Tick the box next to your chosen answer.
4. Section B & C: Write your answers in the spaces provided.
5. Use a blue or black pen. Diagrams may be drawn in pencil.
6. All working must be shown for calculations.
7. The number of marks is given in brackets [] at the end of each question.

Section	Marks Available	Marks Awarded
A	15	
B	20	
C	15	
Total	50	

SECTION A — Multiple Choice [15 marks] Answer all 15 questions. Tick ONE box per question.

- 1 The atomic number of an element is equal to the number of [1]
- ☐ A neutrons in the nucleus
- ☐ B protons in the nucleus
- ☐ C electrons in the outermost shell only
- ☐ D protons plus neutrons in the nucleus
- 2 An atom of oxygen has 8 protons and a mass number of 16. How many neutrons does it contain? [1]
- ☐ A 6
- ☐ B 7
- ☐ C 8
- ☐ D 16
- 3 The electron configuration of sodium (atomic number 11) is [1]
- ☐ A 2, 9
- ☐ B 2, 8, 1
- ☐ C 3, 8
- ☐ D 2, 8, 8, 1
- 4 What is the maximum number of electrons that can occupy the second electron shell? [1]
- ☐ A 2
- ☐ B 8
- ☐ C 18
- ☐ D 32
- 5 Two atoms are isotopes of the same element. They must have the same number of [1]
- ☐ A protons
- ☐ B neutrons
- ☐ C mass numbers
- ☐ D nucleons
- 6 Chlorine has atomic number 17 and electron configuration 2, 8, 7. In which period of the periodic table is chlorine found? [1]

- ☐ A Period 1
- ☐ B Period 2
- ☐ C Period 3
- ☐ D Period 4

7 Elements in Group VII of the periodic table are called

[1]

- ☐ A alkali metals
- ☐ B alkaline earth metals
- ☐ C halogens
- ☐ D noble gases

8 Moving across Period 3 from sodium to chlorine, the atomic radius

[1]

- ☐ A increases steadily
- ☐ B remains constant throughout
- ☐ C decreases steadily
- ☐ D first increases then decreases

9 Going down Group I of the periodic table, the reactivity of the elements

[1]

- ☐ A decreases
- ☐ B increases
- ☐ C remains the same
- ☐ D first increases then decreases

10 Carbon-12 and carbon-14 are isotopes. Compared with carbon-12, carbon-14 has

[1]

- ☐ A 2 fewer protons
- ☐ B 2 more protons
- ☐ C 2 more neutrons
- ☐ D 2 fewer neutrons

11 In the formation of an ionic bond, electrons are

[1]

- ☐ A shared equally between the two atoms
- ☐ B transferred from one atom to another
- ☐ C delocalised throughout the structure
- ☐ D gained by both atoms simultaneously

12 What is the chemical formula of magnesium chloride?

[1]

- ☐ A MgCl

- ☐ B $\text{Mg } 2 \text{ Cl}$
☐ C $\text{MgCl } 2$
☐ D $\text{Mg } 2 \text{ Cl } 3$

13 Calcium has atomic number 20. When a calcium atom loses 2 electrons to form an ion, the correct symbol for this ion is

[1]

- ☐ A $\text{Ca } -$
☐ B $\text{Ca } 2-$
☐ C $\text{Ca } +$
☐ D $\text{Ca } 2+$

14 Noble gases are found in which group of the periodic table?

[1]

- ☐ A Group I
☐ B Group IV
☐ C Group VII
☐ D Group 0

15 Which of the following elements has the electron configuration 2, 8, 8, 2?

[1]

- ☐ A Magnesium (atomic number 12)
☐ B Sulfur (atomic number 16)
☐ C Calcium (atomic number 20)
☐ D Argon (atomic number 18)

SECTION B — Short Structured Questions [20 marks] Answer all questions.

B1 The table below shows data about three particles, X, Y, and Z.

Particle	Protons	Neutrons	Electrons
X	11	12	11
Y	11	13	11
Z	17	18	17

(a) State the atomic number of particle X.

[1]

(b) Particles X and Y are isotopes of the same element. Explain what this means.

[2]

(c) Chlorine exists as two isotopes: ^{35}Cl (75%) and ^{37}Cl (25%). Calculate the relative atomic mass of chlorine. Show your working.

[2]

B2 This question is about trends in the periodic table.

(a) State how first ionisation energy changes across Period 3 from left to right.

[1]

- (b) Explain why atomic radius decreases across Period 3. Use ideas about nuclear charge and electron shells in your answer.

[2]

- (c) Potassium reacts more vigorously with water than sodium does, even though both are in Group I. Explain why.

[2]

B3 This question is about ionic bonding between sodium (Na, atomic number 11) and chlorine (Cl, atomic number 17).

- (a) In the box below, draw a dot-and-cross diagram to show the ionic bond formed between one sodium atom and one chlorine atom. Show **outer-shell electrons only**. Label the ions formed with their charges.

[3]

Draw your diagram here

- (b) State **two** physical properties of sodium chloride that are characteristic of ionic compounds.

[2]

B4 This question is about ions and ionic formulae.

(a) Write the symbol, including charge, of the ion formed when:

(i) oxygen gains 2 electrons

[1]

(ii) aluminium loses 3 electrons

[1]

(b) Use the ions Ca^{2+} , Al^{3+} , O^{2-} , and Cl^- to write the formulae of the following compounds.

(i) Calcium chloride

[1]

(ii) Aluminium oxide

[2]

SECTION C — Long Structured Questions [15 marks] Answer all questions.

C1 Atomic Structure and the Periodic Table

[8]

Lithium (Li), sodium (Na) and potassium (K) are all elements in Group I of the periodic table. The diagram below shows the electron configuration of the sodium atom.

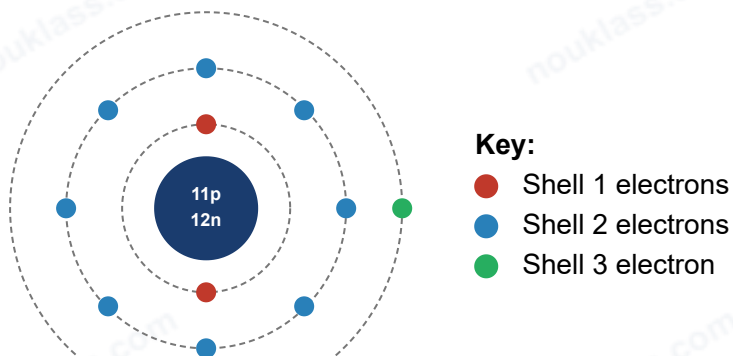


Fig. 1 — Sodium atom (2, 8, 1)

- (a) State what is meant by the term *electron configuration*.

[1]

- (b) In the box below, draw a similar electron configuration diagram for a lithium atom (atomic number 3). Label the nucleus with the correct number of protons and neutrons (lithium-7).

[2]

Draw your diagram here

- (c) Explain why Group I elements become more reactive going down the group from lithium to potassium. Your answer should refer to electron shells, nuclear charge, and ease of electron loss.

[3]

- (d) Write the **word equation** for the reaction of potassium with water.

[1]

- (e) State what you would observe if a small piece of potassium is placed into water. Give **one** observation that is different from the observation when sodium reacts with water.

[1]

C2 Ionic Bonding and Properties of Ionic Compounds

[7]

Magnesium oxide (MgO) is an ionic compound widely used as a refractory lining in furnaces because it withstands very high temperatures.

- (a) Magnesium has atomic number 12. Write the electron configuration of a magnesium atom.

[1]

- (b) In the box below, draw a dot-and-cross diagram to show the ionic bonding in magnesium oxide. Show **outer-shell electrons only**. Label the ions with their charges.

[3]

Draw your diagram here

- (c) Magnesium oxide has a higher melting point than sodium chloride. Explain why, in terms of ionic charges and electrostatic attraction.

[2]

- (d) Solid magnesium oxide does not conduct electricity, but molten magnesium oxide does. Explain this difference.

[1]

Mark Scheme — Paper 1

For teacher/self-assessment use only. Accept any answer that conveys the correct scientific meaning. Credit alternative correct responses. Marks in brackets indicate the allocation per point. Underlined words are essential for credit.

Section A — Multiple Choice

Q	Answer	Q	Answer	Q	Answer
1	B	6	C	11	B
2	C	7	C	12	C
3	B	8	C	13	D
4	B	9	B	14	D
5	A	10	C	15	C

Section B

Question	Mark-worthy points	Marks
B1(a)	11	1
B1(b)	Same number of <u>protons</u> [1]; different number of <u>neutrons</u> [1]	2
B1(c)	Working: $(35 \times 75 + 37 \times 25) \div 100$ [1]; Answer: 35.5 [1]	2
B2(a)	First ionisation energy <u>increases</u> across Period 3 (left to right)	1
B2(b)	Number of protons (nuclear charge) increases [1]; electrons added to <u>same shell</u> , so increased attraction pulls electrons closer to nucleus [1]	2
B2(c)	Potassium has one more electron shell than sodium [1]; outer electron is further from the nucleus / greater shielding, so it is lost more easily [1]	2
B3(a)	Na shown with 1 outer electron [1]; Cl shown with 7 outer electrons [1]; after transfer: Na^+ (empty outer shell) and Cl^- (8 outer electrons) correctly labelled with charges [1]	3
B3(b)	Any 2 from: high melting/boiling point; conducts electricity when molten or dissolved in water; brittle; soluble in water	2
B4(a)(i)	O^{2-}	1
B4(a)(ii)	Al^{3+}	1
B4(b)(i)	CaCl_2	1
B4(b)(ii)	Al_2O_3 [1 for correct elements, 1 for correct subscripts]	2

Section C

Question	Mark-worthy points	Marks
C1(a)	The arrangement/distribution of electrons in shells/energy levels around the nucleus	1
C1(b)	Nucleus labelled 3p, 4n [1]; 2 electrons in shell 1, 1 electron in shell 2 correctly drawn [1]	2
C1(c)	Going down Group I, each element has one more electron shell [1]; outer electron is further from the nucleus [1]; greater shielding by inner electrons, nuclear attraction on outer electron is weaker [1]; outer electron lost more easily, so reactivity increases [bonus point — award 3rd mark for this link]	3
C1(d)	Potassium + water → potassium hydroxide + hydrogen	1
C1(e)	Potassium floats / moves rapidly on surface / fizzes / melts into a ball; difference: potassium <u>ignites / burns with a lilac/pink flame</u> (sodium does not)	1
C2(a)	2, 8, 2	1
C2(b)	Mg shown with 2 outer electrons [1]; O shown with 6 outer electrons [1]; after transfer: Mg^{2+} (empty outer shell) and O^{2-} (8 outer electrons) with correct charges [1]	3
C2(c)	Mg^{2+} and O^{2-} ions carry <u>higher charges (2+/2-)</u> than Na^+/Cl^- [1]; stronger electrostatic attraction between ions / more energy needed to break the lattice, so higher melting point [1]	2
C2(d)	In solid MgO, ions are in fixed positions / cannot move; when molten, ions are free to move and carry charge	1